AP BIOLOGY TOPIC 5.5

Environmental Effects on Phenotype & Phenotypic Plasticity

UNIT 5: HEREDITY (8–11% EXAM WEIGHT)

SYI-3.B | PHENOTYPIC PLASTICITY | GENE × ENVIRONMENT | NORMS OF REACTION

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 5.5 (ENVIRONMENTAL EFFECTS ON PHENOTYPE & PHENOTYPIC PLASTICITY)

CED Core Vocabulary

Definition — Phenotypic plasticity occurs when individuals with the same genotype exhibit different phenotypes when exposed to different environmental conditions.

Norm of Reaction — A graph plotting the range of phenotypes produced by a single genotype across a continuous environmental gradient (e.g., plant height across elevation or...

Enhanced Survival & Fitness — Organisms achieve higher reproductive success across heterogeneous or unpredictable habitats.

Metabolic Energy Cost — Maintaining sensory receptors and regulatory machinery for flexible gene expression requires more energy than a genetically fixed phenotype.

Molecular Mechanism — Environmental signals alter gene expression (transcription factor activity, epigenetic chromatin modification) or protein/enzyme activity — does NOT alter the...

Rapid Adaptability Within One Lifetime — Allows an individual organism to respond to fluctuating environmental conditions immediately without waiting for slow generational changes in allele frequencies.

Resource & Energy Efficiency — Allocates metabolic energy to inducible traits (e.g., chemical defenses or protective structures) only when needed , avoiding the cost of maintaining permanent...

Physiological Trade-Offs — Enhancing one trait often comes at the expense of another (e.g., shade plants elongating stems rapidly toward light develop weaker root systems and thinner stems).

§2. PHENOTYPIC PLASTICITY: SAME GENOTYPE, DIFFERENT PHENOTYPE

Gene Expression Shift

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Phenotypic Plasticity ➔ **Definition** ➔ **Molecular Mechanism** ➔ **Norm of Reaction**

- **Definition:** **Phenotypic plasticity** occurs when individuals with the **same genotype** exhibit **different phenotypes** when exposed to different environmental conditions.
- **Molecular Mechanism:** Environmental signals alter **gene expression** (transcription factor activity, epigenetic chromatin modification) or **protein/enzyme activity** — does **NOT** alter the underlying DNA nucleotide sequence!
- **Norm of Reaction:** A graph plotting the range of phenotypes produced by a single genotype across a continuous environmental gradient (e.g., plant height across elevation or temperature).

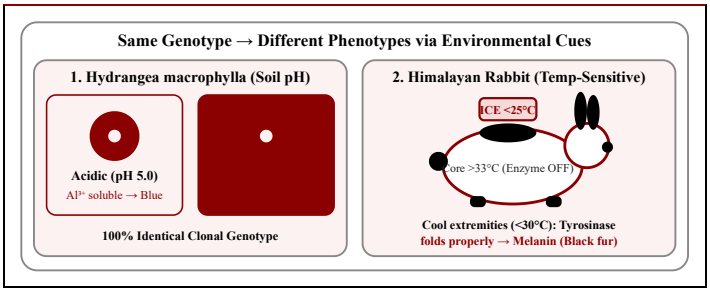
[MECHANISTIC NOTE] **Genotype Stays Constant:** Only the *level of transcription/translation* or *enzyme functionality* changes in response to the environment.

[CORE AXIOM / REMEMBER] **Analogy:** Think of the genotype as the same **"recipe"** producing a different **"dish"** depending on oven conditions — or the environment acting as a **"dimmer switch"** for genes!

[COMMON AP EXAM FRQ TRAP] **Plasticity ≠ Mutation:** Never say an environmental factor "mutated the gene to fit the environment" in phenotypic plasticity — **no DNA sequence change occurs**!

§4. VISUALIZING PHENOTYPIC PLASTICITY MECHANISMS

Enzyme & Soil pH Models



§5. EVOLUTIONARY ADVANTAGES & LIMITATIONS OF PLASTICITY

Fitness & Trade-Offs

- **Evolutionary Advantages of Phenotypic Plasticity:**
  - **Rapid Adaptability Within One Lifetime:** Allows an individual organism to respond to fluctuating environmental conditions immediately without waiting for slow generational changes in allele frequencies.
  - **Enhanced Survival & Fitness:** Organisms achieve higher reproductive success across heterogeneous or unpredictable habitats.
  - **Resource & Energy Efficiency:** Allocates metabolic energy to inducible traits (e.g., chemical defenses or protective structures) *only when needed*, avoiding the cost of maintaining permanent structures.
- **Limitations & Evolutionary Costs of Plasticity:**
  - **Metabolic Energy Cost:** Maintaining sensory receptors and regulatory machinery for flexible gene expression requires more energy than a genetically fixed phenotype.
  - **Physiological Trade-Offs:** Enhancing one trait often comes at the expense of another (e.g., shade plants elongating stems rapidly toward light develop weaker root systems and thinner stems).
  - **Environmental Mismatch Risk:** If environmental cues are unreliable or change faster than the plastic response time, the induced phenotype may become maladaptive.

[EXAMPLE & BIOLOGICAL APPLICATION] **Desert Plants & Water Stress:** Many desert shrubs shed leaves or develop smaller, thicker, waxier leaves during drought seasons to minimize stomatal transpiration water loss, then grow larger photosynthetic leaves after seasonal rains.

[COMMON AP EXAM FRQ TRAP] **Plasticity Has Costs & Limits:** Do not assume phenotypic plasticity is always beneficial or limitless — extreme environments beyond a species' physiological tolerance still cause death, and plastic traits are constrained by energetic trade-offs.

§3. HIGH-YIELD AP EXAMPLES OF GENE × ENVIRONMENT INTERACTION

Must-Know Classics

ENVIRONMENTAL FACTOR	ORGANISM / TRAIT	PHYSIOLOGICAL / MOLECULAR MECHANISM
Soil pH & Aluminum (Al^{3+})	**Hydrangea Flower Color**	Same genotype produces **blue flowers in acidic soil (pH < 5.5)** (free Al^{3+} ions are soluble and absorbed, complexing with anthocyanin pigment) vs. **pink flowers in alkaline/neutral soil** (Al^{3+} precipitates out and is unavailable).
Temperature (Enzyme Folding)	**Himalayan Rabbit / Siamese Cat Coat**	Homozygous for a **heat-sensitive tyrosinase enzyme** (synthesizes dark melanin). Enzyme denatures at warm core body temp (>33°C → **white fur**) but remains folded & active in cooler extremities (ears, nose, paws, tail → **black fur**).
Incubation Temperature	**Reptile Sex Determination (TSD)**	In sea turtles and alligators, egg incubation temperature during a thermosensitive embryonic window regulates *aromatase* gene expression (converting androgens to estrogens) → determines male vs. female differentiation.
UV Light / Photoperiod	**Human Melanin & Arctic Fox / Ptarmigan**	UV radiation stimulates melanocytes to upregulate *tyrosinase* transcription (skin tanning). Seasonal photoperiod/temp shifts trigger white winter vs. brown summer coats in arctic foxes.
Population Density / Diet	**Locusts, Honeybees & *Daphnia***	Crowding switches desert locusts from solitary green to gregarious swarming phase; royal jelly diet methylates DNA in bee larvae to develop into fertile Queens vs. sterile Workers; predator kairomones induce protective "helmets" in *Daphnia*.



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§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 5.5 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 5.5: Environmental Effects on Phenotype & Phenotypic Plasticity ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 $SE_{\bar{x}}$)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 7 Part 2 (Sordaria fimicola Meiotic Crossing Over) & Drosophila/Fast Plants Crosses: Counting 4:4 non-recombinant vs. 2:2:2:2 / 2:4:2 recombinant asci in Sordaria perithecia to determine gene-to-centromere map distance (% Crossover = [Recombinant asci / Total asci] × 100; Map distance in cM = % Crossover / 2 because only 2 of 4 chromatids in a tetrad cross over). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Statistical Testing: Chi-Square Goodness-of-Fit: $\chi^2 = \Sigma [(O - E)^2 / E]$, with Degrees of Freedom df = (number of phenotypic classes − 1). At $\alpha = 0.05$: if $\chi^2 >$ critical value, REJECT H_0 (e.g., genes are autosomal linked rather than assorting independently 9:3:3:1); if $\chi^2 \leq$ critical value, FAIL TO REJECT H_0. Recombination Frequency: RF = (Total Recombinants / Total Offspring) × 100% = Map Units (cM). **Error Bar Rule (±2 $SE_{\bar{x}}$):** If the ±2 $SE_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Chromosomal Disruption Case Study: Meiotic Nondisjunction (failure of homologous chromosomes to separate in Anaphase I, or sister chromatids to separate in Anaphase II) yields aneuploid gametes (n + 1 or n − 1). Fertilization produces Trisomy 21 (Down Syndrome, 47, +21), Klinefelter Syndrome (47, XXY), or monosomic Turner Syndrome (45, X0). Mitochondrial myopathies (e.g., Leber Hereditary Optic Neuropathy) exhibit strict maternal inheritance because the ovum contributes virtually all zygotic cytoplasm and mitochondria.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 5.5 to prove why the claim holds.